RITHUL KOSAL
251901135

CRYPTOGRAPHY

EXP NO: 1

ENCRYPTION - CRYPTO 101

AIM: To learn the importance of cryptography and different encryption techniques and introduction to encryption.

TASK 2 :

- Are SSH keys protected with a passphrase or a password?
  → Passphrase

TASK 3 :

- What does SSH stand for?
  → Secure Shell
- How do web servers prove their identity?
  → Certificates
- What is the main set of standards you need to comply with if you store or process payment card details?
  → PCI-DSS

TASK 4 :

- What's 30%5?
  → 0
- What's 25%7?
  → 4
- What's 118613842%9091?
  → 3565

TASK 5 :

- Should you trust DES? Yea/Nay
  → Nay
- What was the result of the attempt to make DES more secure so that it could be used for longer?
  → Triple DES
- Is it okay to share your public key? Yea/Nay
  → Yea

TASK 6 :

- p= 4391, q= 6659, What is n?
  → 29239669

TASK 8 :

- What can you use to verify that a file has not been modified and is the authentic file as the author needed?
  → Digital Systems

TASK 9:

- What algorithm does the key use?
  → RSA
- Crack the password with the “John The Ripper” and “rockyou”, What’s the passphrase for key?
  → DELICIOUS

TASK 11 :

-
- You have the private key and a file encrypted with the public key, decrypting the file. What's the secret word?
  → Pineapple

Room completed: 100%

DO NOT encrypt passwords unless you're doing something like a password manager. Passwords should not be stored in plaintext, and you should use hashing to manage them safely.

Answer the questions below

What does SSH stand for?

Secure Shell ✓ Correct Answer

How do webservers prove their identity?

certificates ✓ Correct Answer

What is the main set of standards you need to comply with if you store or process payment card details?

PCI-DSS ✓ Correct Answer

Room completed: 100%

RSA and Elliptic Curve cryptography are based around different mathematically difficult (intractable) problems, which give them their strength. More about RSA later.

Answer the questions below

Should you trust DES? Yea/Nay

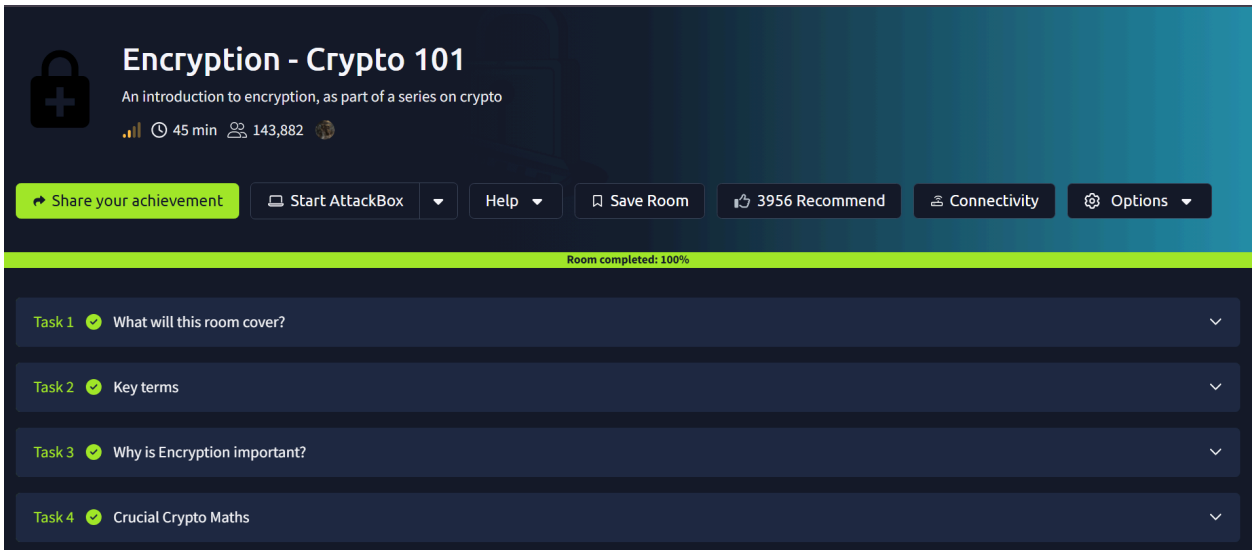
Nay ✓ Correct Answer

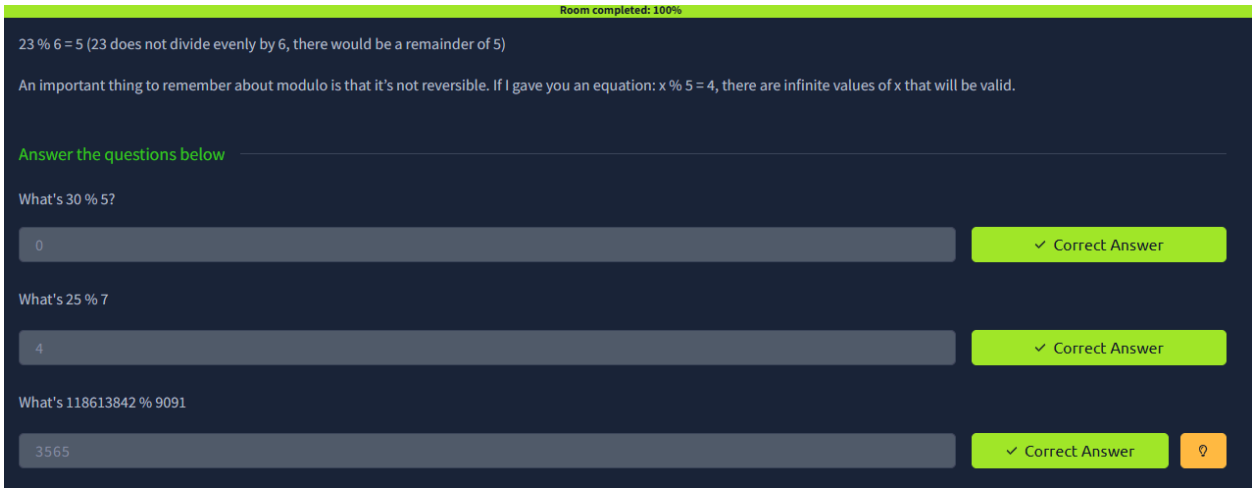
What was the result of the attempt to make DES more secure so that it could be used for longer?

Triple DES ✓ Correct Answer

Is it ok to share your public key? Yea/Nay

Yea ✓ Correct Answer





RESULT: Thus different encryption techniques and introduction to cryptography has been studied.